

Follow today's class instructions (dbt only. No need to run it as a DAG)

1. Create a dbt project with snowflake connector (1 pt)

dbt init HW4

```
(data226_dbt) C:\Users\kevin\OneDrive\桌面\DATA226>dbt init HW4
07:36:49 Running with dbt=1.12.5
07:36:49 Setting up your profile.
Which database would you like to use?
[1] snowflake

(Don't see the one you want? https://docs.getdbt.com/docs/available-adapters)

Enter a number: 1
account (https://<this_value>.snowflakecomputing.com): IKFRRDP-BK44433
user (dev username): CHINGTU
[1] password
[2] keypair
[3] sso
[4] workload_identity
Desired authentication type option (enter a number): 2
private_key_path (path/to/private.key): C:\Users\kevin\OneDrive\桌面\DATA226\sjsu-data226-FA26\keys\rsa_key.p8
private_key_passphrase (passphrase for the private key, if key is encrypted):
role (dev role): ACCOUNTADMIN
warehouse (warehouse name): CTDATA226
database (default database that dbt will build objects in): HW4
schema (default schema that dbt will build objects in): raw
threads (1 or more) [1]: 1
07:37:57 Profile HW4 written to C:\Users\kevin\.dbt\profiles.yml using target's profile_template.yml and your supplied values.
07:37:57 Running dbt debug to validate the project...
07:37:57 dbt version: 1.12.5
07:37:57 python version: 3.12.14
07:37:57 python path: C:\Users\kevin\anaconda3\envs\data226_dbt\python.exe
07:37:57 os info: Windows-11-10.0.26200-SP0
07:37:57 Using profiles dir at C:\Users\kevin\.dbt\profiles.yml
07:37:57 Using dbt_project.yml file at C:\Users\kevin\OneDrive\桌面\DATA226\HW4\dbt_project.yml
07:37:57 adapter type: snowflake
07:37:57 adapter version: 1.12.1
07:37:57 Configuration:
07:37:57   profiles.yml file [OK found and valid]
07:37:57   dbt_project.yml file [OK found and valid]
07:37:57 Required dependencies:
07:37:57   - git [OK found]

07:37:57 Connection:
07:37:57   account: IKFRRDP-BK44433
07:37:57   user: CHINGTU
07:37:57   database: HW4
07:37:57   warehouse: CTDATA226
07:37:57   role: ACCOUNTADMIN
07:37:57   schema: raw
07:37:57   authenticator: None
07:37:57   oauth_client_id: None
07:37:57   query_tag: None
07:37:57   client_session_keep_alive: False
07:37:57   host: None
07:37:57   port: None
07:37:57   proxy_host: None
07:37:57   proxy_port: None
07:37:57   protocol: None
07:37:57   connect_retries: 1
07:37:57   connect_timeout: None
07:37:57   retry_on_database_errors: False
07:37:57   retry_all: False
07:37:57   insecure_mode: False
07:37:57   reuse_connections: True
07:37:57   s3_stage_vpce_dns_name: None
07:37:57   workload_identity_provider: None
07:37:57   workload_identityentra_resource: None
07:37:57   platform_detection_timeout_seconds: 0.0
07:37:57 Registered adapter: snowflake=1.12.1
07:37:58 Connection test: [OK connection ok]

07:37:58 All checks passed!
07:37:58 Your new dbt project "HW4" was created!
```

dbt debug

```
(data226_dbt) C:\Users\kevin\OneDrive\桌面\DATA226\HW4>dbt debug
07:38:28 Running with dbt=1.12.5
07:38:28 dbt version: 1.12.5
07:38:28 python version: 3.12.14
07:38:28 python path: C:\Users\kevin\anaconda3\envs\data226_dbt\python.exe
07:38:28 os info: Windows-11-10.0.26200-SP0
07:38:29 Using profiles dir at C:\Users\kevin\.dbt
07:38:29 Using profiles.yml file at C:\Users\kevin\.dbt\profiles.yml
07:38:29 Using dbt_project.yml file at C:\Users\kevin\OneDrive\桌面\DATA226\HW4\dbt_project.yml
07:38:29 adapter type: snowflake
07:38:29 adapter version: 1.12.1
07:38:29 Configuration:
07:38:29   profiles.yml file [OK found and valid]
07:38:29   dbt_project.yml file [OK found and valid]
07:38:29 Required dependencies:
07:38:29   - git [OK found]

07:38:29 Connection:
07:38:29   account: IKFRRDP-BK44433
07:38:29   user: CHINGTU
07:38:29   database: HW4
07:38:29   warehouse: CTDATA226
07:38:29   role: ACCOUNTADMIN
07:38:29   schema: raw
07:38:29   authenticator: None
07:38:29   oauth_client_id: None
07:38:29   query_tag: None
07:38:29   client_session_keep_alive: False
07:38:29   host: None
07:38:29   port: None
07:38:29   proxy_host: None
07:38:29   proxy_port: None
07:38:29   protocol: None
07:38:29   connect_retries: 1
07:38:29   connect_timeout: None
07:38:29   retry_on_database_errors: False
07:38:29   retry_all: False
07:38:29   insecure_mode: False
07:38:29   reuse_connections: True
07:38:29   s3_stage_vpce_dns_name: None
07:38:29   workload_identity_provider: None
07:38:29   workload_identityentra_resource: None
07:38:29   platform_detection_timeout_seconds: 0.0
07:38:29 Registered adapter: snowflake=1.12.1
07:38:48 Connection test: [OK connection ok]
```

2. Set up transform models (2pt)

Dummy data build using snowflake

The screenshot displays the Snowflake SQL editor interface. The left pane shows the execution of SQL commands to create two tables and insert data. The right pane shows the results of the insert operations, indicating that 8 rows were inserted into each table.

```
USE ROLE ACCOUNTADMIN;
USE DATABASE HW4;
CREATE SCHEMA IF NOT EXISTS RAW;

CREATE OR REPLACE TABLE RAW.USER_SESSION_CHANNEL(
  userid INT,
  sessionid VARCHAR(32) PRIMARY KEY,
  channel VARCHAR(32)
);

INSERT INTO RAW.USER_SESSION_CHANNEL (userid, sessionid, channel) VALUES
(1, 's1', 'facebook'),
(2, 's2', 'google'),
(3, 's3', 'facebook'),
(4, 's4', 'youtube'),
(5, 's5', 'google'),
(6, 's6', 'instagram'),
(7, 's7', 'facebook'),
(8, 's8', 'youtube');

CREATE OR REPLACE TABLE RAW.SESSION_TIMESTAMP(
  sessionid VARCHAR(32) PRIMARY KEY,
  ts TIMESTAMP
);

INSERT INTO RAW.SESSION_TIMESTAMP(sessionid, ts) VALUES
('s1', '2026-09-01 10:00:00'),
('s2', '2026-09-01 11:15:00'),
('s3', '2026-09-02 09:30:00'),
('s4', '2026-09-02 14:45:00'),
('s5', '2026-09-03 08:20:00'),
('s6', '2026-09-03 16:10:00'),
('s7', '2026-09-04 12:00:00'),
('s8', '2026-09-04 18:30:00');
```

Query Results (just now):

#	number of rows inserted
1	8

#	number of rows inserted
1	8

Query Results (just now):

#	USERID	SESSIONID	CHANNEL
1	1	s1	facebook
2	2	s2	google
3	3	s3	facebook
4	4	s4	youtube
5	5	s5	google
6	6	s6	instagram
7	7	s7	facebook
8	8	s8	youtube

Query Results (just now):

#	SESSIONID	TS
1	s1	2026-09-01 10:00:00.000
2	s2	2026-09-01 11:15:00.000
3	s3	2026-09-02 09:30:00.000
4	s4	2026-09-02 14:45:00.000
5	s5	2026-09-03 08:20:00.000
6	s6	2026-09-03 16:10:00.000
7	s7	2026-09-04 12:00:00.000
8	s8	2026-09-04 18:30:00.000

Making folder of transform and analytics

The screenshot shows a Windows command prompt window with the following commands and output:

```
(data226_dbt) C:\Users\kevin\OneDrive\桌面\DATA226\HW4>ls
README.md analyses dbt_project.yml logs macros models seeds snapshots tests

(data226_dbt) C:\Users\kevin\OneDrive\桌面\DATA226\HW4>mkdir models\transform

(data226_dbt) C:\Users\kevin\OneDrive\桌面\DATA226\HW4>mkdir models\analytics
```

Below the command prompt, a screenshot of the OneDrive file explorer shows the 'models' folder containing the newly created 'analytics' and 'transform' subfolders.

名稱	狀態	修改日期	類型	大小
analytics	✓	2026/9/18 上午 12:55	檔案資料夾	
transform	✓	2026/9/18 上午 12:54	檔案資料夾	

1. Input (transform) tables should be built as **CTE** (involves changes to dbt_project.yml)
2. Show contents of models/transform/user_session_channel.sql and transform/session_timestamp.sql

models\source.yml and dbt_project.yml

```
version: 2

sources:
  - name: raw
    database: HW4
    schema: raw
    description: "Schema which stores raw tables (ETL)"
    tables:
      - name: user_session_channel
        identifier: user_session_channel
        description: "Session data including userId, sessionId, and channel"
      - name: session_timestamp
        identifier: session_timestamp
        description: "Session data including timestamp"

models:
  HW4:
    # Config indicated by + and
    transform:
      +materialized: ephemeral
    analytics:
      +materialized: table
```

models\transform\user_session_channel.sql

```
with source as (
  select
    userId,
    sessionId,
    channel
  from {{ source('raw', 'user_session_channel') }}
)

select *
from source
WHERE sessionId IS NOT NULL
```

models\transform\session_timestamp.sql

```
with source as (
  select sessionId, ts
  from {{ source('raw', 'session_timestamp') }}
)

select * from source
WHERE sessionId IS NOT NULL
```

dbt run (Note: I set ephemeral to transform models to output CTE instead of view)

```
(data226-dbt) C:\Users\Kevin\OneDrive\桌面\DATA226\HW4>dbt run
21:47:25 Running with dbt=1.12.5
21:47:26 Registered adapter: snowflake=1.12.1
21:47:26 Unable to do partial parsing because saved manifest not found. Starting full parse.
21:47:27 [WARNING]: Did not find matching node for patch with name 'session_summary' in the 'models' section of file 'models/schema.yml'
21:47:27 [WARNING]: Test 'test.HW4.unique_session_summary_sessionId.9c5e5661a0' (models/schema.yml) depends on a node named 'session_summary' in package '' which was not found
21:47:27 [WARNING]: Test 'test.HW4.not_null_session_summary_sessionId.e6aecla6c6' (models/schema.yml) depends on a node named 'session_summary' in package '' which was not found
21:47:27 [WARNING]: Configuration paths exist in your dbt_project.yml file which do not apply to any resources.
There are 1 unused configuration paths:
- models.HW4.analytics
21:47:27 Found 2 models, 2 sources, 562 macros
21:47:27 Concurrency: 1 threads (target='dev')
21:47:27
21:47:27 Finished running in 0 hours 0 minutes and 0.02 seconds (0.02s).
```

3. Set up analytics models (1pt)

1. Show contents of models/analytics/session_summary.sql

models\analytics\session_summary.sql

```
with user_session as (  
    select * from {{ref('user_session_channel')}}  
),  
  
session_ts as (  
    select * from {{ref('session_timestamp')}}  
),  
  
final as (  
    select  
        u.userId,  
        u.sessionId,  
        u.channel,  
        s.ts  
    from user_session u  
    join session_ts s on u.sessionId = s.sessionId  
)  
select * from final
```

dbt run

```
(data226_dbt) C:\Users\kevin\OneDrive\桌面\DATA226\HW4>dbt run  
22:08:34 Running with dbt=1.12.5  
22:08:35 Registered adapter: snowflake=1.12.1  
22:08:35 Found 3 models, 2 sources, 562 macros  
22:08:35 Concurrency: 1 threads (target='dev')  
22:08:35 1 of 1 START sql table model raw.session_summary ..... [RUN]  
22:08:40 1 of 1 OK created sql table model raw.session_summary ..... [SUCCESS 8 in 2.56s]  
22:08:46 Finished running 1 table model in 0 hours 0 minutes and 10.19 seconds (10.19s).  
22:08:46 Completed successfully  
22:08:46 Done. PASS=1 WARN=0 ERROR=0 SKIP=0 NO-OP=0 REUSED=0 TOTAL=1
```

Result table on snowflake

SELECT * FROM RAW.SESSION_SUMMARY;				
results (8 minutes ago)				
	USERID	SESSIONID	CHANNEL	TS
1	1	s1	facebook	2026-09-01 10:00:00.000
2	2	s2	google	2026-09-01 11:15:00.000
3	3	s3	facebook	2026-09-02 09:30:00.000
4	4	s4	youtube	2026-09-02 14:45:00.000
5	5	s5	google	2026-09-03 08:20:00.000
6	6	s6	instagram	2026-09-03 16:10:00.000
7	7	s7	facebook	2026-09-04 12:00:00.000
8	8	s8	youtube	2026-09-04 18:30:00.000

4. Add Snapshot to the analytics table (session_summary) (2pt)

1. Show contents of snapshots/snapshot_session_summary.sql

snapshots\snapshots_session_summary.sql

```
{% snapshot snapshot_session_summary %}

{{
    config(
        target_database='HW4',
        target_schema='analytics',
        unique_key='sessionId',
        strategy='timestamp',
        updated_at='ts'
    )
}}

select *
from {{ ref('session_summary') }}

{% endsnapshot %}
```

dbt snapshot

```
(data226_dbt) C:\Users\kevin\OneDrive\桌面\DATA226\HW4>dbt snapshot
22:19:35 Running with dbt=1.12.5
22:19:36 Registered adapter: snowflake=1.12.1
22:19:37 Found 3 models, 1 snapshot, 2 sources, 562 macros
22:19:37 Concurrency: 1 threads (target='dev')
22:19:37 1 of 1 START snapshot analytics.snapshot_session_summary ..... [RUN]
22:19:40 1 of 1 OK snapshot analytics.snapshot_session_summary ..... [SUCCESS 8 in 2.49s]
22:19:43 Finished running 1 snapshot in 0 hours 0 minutes and 6.56 seconds (6.56s).
22:19:44 Completed successfully
22:19:44 Done. PASS=1 WARN=0 ERROR=0 SKIP=0 NO-OP=0 REUSED=0 TOTAL=1
```

Result table on snowflake

SELECT FROM HW4.ANALYTICS.SNAPSHOT_SESSION_SUMMARY;									
Results (2 minutes ago)									
ID	USERID	SESSIONID	CHANNEL	TS	DBT_SCD_ID	DBT_UPDATED_AT	DBT_VALID_FROM	DBT_VALID_TO	
1	1	s1	facebook	2026-09-01 10:00:00.000	9c115f67d200617b8f6c9de5590ef9d	2026-09-01 10:00:00.000	2026-09-01 10:00:00.000	null	
2	2	s2	google	2026-09-01 11:15:00.000	8a01f26721d572b3b2b07748530a310	2026-09-01 11:15:00.000	2026-09-01 11:15:00.000	null	
3	3	s3	facebook	2026-09-02 09:30:00.000	2b89b30358ec244e1391f8b7d8e981d2	2026-09-02 09:30:00.000	2026-09-02 09:30:00.000	null	
4	4	s4	youtube	2026-09-02 14:45:00.000	e9213b647227718b9665f25a221c80	2026-09-02 14:45:00.000	2026-09-02 14:45:00.000	null	
5	5	s5	google	2026-09-03 08:20:00.000	77c8f1b2f8fb9b04c57b3ac7567d7b7	2026-09-03 08:20:00.000	2026-09-03 08:20:00.000	null	
6	6	s6	instagram	2026-09-03 16:10:00.000	e0b15444abab3159c8bf9ecf55351a3b	2026-09-03 16:10:00.000	2026-09-03 16:10:00.000	null	
7	7	s7	facebook	2026-09-04 12:00:00.000	4fb78999f9fb25d82189d5e52b314394	2026-09-04 12:00:00.000	2026-09-04 12:00:00.000	null	
8	8	s8	youtube	2026-09-04 18:30:00.000	fee0e9a3f6f794baffa10bf4541964b	2026-09-04 18:30:00.000	2026-09-04 18:30:00.000	null	

5. Add at least 2 tests to sessionId field of session_summary (2pt)

1. Show contents of models/schema.yml

models/schema.yml

```
version: 2

models:
  - name: session_summary
    description: "Analytics model for session data"
    columns:
      - name: SESSIONID
        description: "Unique identifier for each session"
        data_tests:
          - unique
          - not_null
```

dbt test

```
(data226_dbt) C:\Users\kevin\OneDrive\桌面\DATA226\HW4>dbt test
22:35:12 Running with dbt=1.12.5
22:35:13 Registered adapter: snowflake=1.12.1
22:35:13 Found 3 models, 1 snapshot, 2 data tests, 2 sources, 562 macros
22:35:13 Concurrency: 1 threads (target='dev')
22:35:13
22:35:15 1 of 2 START test not_null_session_summary_SESSIONID ..... [RUN]
22:35:17 1 of 2 PASS not_null_session_summary_SESSIONID ..... [PASS in 1.32s]
22:35:17 2 of 2 START test unique_session_summary_SESSIONID ..... [RUN]
22:35:17 2 of 2 PASS unique_session_summary_SESSIONID ..... [PASS in 0.37s]
22:35:17
22:35:17 Finished running 2 data tests in 0 hours 0 minutes and 3.95 seconds (3.95s).
22:35:17 Completed successfully
22:35:17 Done. PASS=2 WARN=0 ERROR=0 SKIP=0 NO-OP=0 REUSED=0 TOTAL=2
```

- For each step, capture screenshots accordingly (examples to follow)
- Create a github repo for this and add the link to the report

https://github.com/ching-tuhuang/data226_FAL26_HW4.git